

Shivam Kumar

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Summary

AI Full Stack Developer with hands-on experience building end-to-end web applications and ML-integrated APIs. Proficient in the MERN stack, FastAPI, and Python ML frameworks (scikit-learn, TensorFlow, OpenCV). Proven ability to ship production-ready systems — real-time web apps, computer vision inference APIs, and NLP pipelines. Pursuing B.Tech in Computer Science at IIIT Kottayam (Expected: May 2027).

Education

Indian Institute of Information Technology, Kottayam • B.Tech in Computer Science *Aug 2023 – May 2027 (Expected)*

- Relevant coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Computer Networks

Technical Skills

Full Stack: React.js, Next.js, Node.js, Express.js, FastAPI, REST APIs, JWT Auth, Framer Motion

Languages: Python, JavaScript, C++, SQL

Databases: MongoDB, MySQL

AI / ML: scikit-learn, TensorFlow, OpenCV, YOLO, NLTK, SpaCy — NLP & Computer Vision pipelines

Cloud & Tools: Git, GitHub, Docker (basic), Google Colab, Jupyter Notebook, Cloudinary, Postman

Experience

Full Stack Developer Intern • kognitiminds Pvt. Ltd. — Remote, India *Jul 2025 – Dec 2025*

- Built and deployed a production-ready school management platform using the **MERN stack**, serving student, teacher, and admin workflows.
- Designed responsive **React.js UI with Framer Motion** animations, improving usability and engagement.
- Architected **RESTful APIs** for core business flows; implemented **JWT-based authentication** and role-based authorization.
- Managed MongoDB database operations ensuring API reliability and data integrity across iterative feature releases.

AI/ML Trainee • Infosys Springboard 7 — AI/ML Track *Jan 2026 – Present*

- Hands-on training in **NLP, Computer Vision, and Deep Learning** pipelines using TensorFlow, PyTorch, and scikit-learn.
- Building and evaluating ML models on real-world datasets; applying accuracy, precision, recall, and F1-score metrics.

Projects

AI Object Detection Web App

YOLO · FastAPI · React.js · OpenCV

2026 march

[GitHub](#) | [Live Demo](#)

- Built an **end-to-end AI Full Stack app** — YOLO model served via FastAPI inference API with a React.js dashboard for real-time image and video object detection.
- **Sub-200ms inference latency** per frame; modular backend supports multiple YOLO model variants via REST endpoints.
- Optimized video processing pipeline with **OpenCV** enabling batch inference and live-stream support.
- Deployed on Render with environment-based config for production and development modes.

Sentiment Analysis System

Python · scikit-learn · FastAPI · React.js

2026 jan

[GitHub](#) | [Live Demo](#)

- End-to-end NLP web app: ML classifier with **89%+ accuracy**, exposed via FastAPI, with a React frontend for real-time text classification.
- Built preprocessing pipelines — tokenization, stopwords removal, TF-IDF vectorization — as reusable Python modules.
- Benchmarked Logistic Regression, Naive Bayes, and SVM; selected best model on F1-score.

Full Stack Chat Application

2025 dec

React.js · Node.js · Express.js · MongoDB

[GitHub](#) | [Live Demo](#)

- Built a real-time chat application with secure **JWT authentication**, user presence tracking, and instant messaging features.
- Designed scalable backend architecture capable of supporting high concurrent user traffic through modular API and database design.
- Optimized frontend state management and backend communication to provide low-latency message delivery with minimal lag.
- Implemented secure session handling, protected routes, and authentication middleware to enhance application security.

Microservice Service Hub

2025 oct

MERN · Microservices · Cloudinary · JWT

[GitHub](#) | [Live Demo](#)

- Developed a scalable **microservices-based platform** with independent services for authentication, media management, user operations, and business workflows.
- Implemented **Role-Based Access Control (RBAC)** supporting Admin, Service Provider, and Customer roles with secure authorization policies.
- Designed secure API communication using JWT authentication and middleware-based authorization mechanisms.
- Integrated Cloudinary for image storage, optimization, and secure media delivery.
- Architected services for scalability, fault isolation, maintainability, and reliable deployment across multiple modules.

Certifications

- **Data Structures and Algorithms** — Udemy (2025)
- **AI & Machine Learning Bootcamp** — Udemy (2025)
- **Complete Full Stack Web Development Bootcamp** — Angela Yu, Udemy (2025)
- **Machine Learning Specialization** — Andrew Ng, Coursera (2024)